

DayFlow AI — Technical Pitch Deck & Architecture Overview

Intelligent Human Resource Management System (HRMS)

1. Executive Summary & Problem Statement

DayFlow AI is an enterprise-grade, full-stack Human Resource Management System engineered to digitize core workforce operations. The system unifies employee self-service, shift attendance tracking, atomic leave management workflows, multi-structured payroll generation, real-time notification streaming, and grounded artificial intelligence policy assistance powered by Google Gemini AI.

2. Technical Stack and Infrastructure Architecture

Layer	Technologies & Frameworks	Key Capabilities
Backend API	Python 3.10+, FastAPI, AsyncIO, Uvicorn	Asynchronous REST pipeline, high-concurrency request routing
Database & ORM	PostgreSQL (asyncpg), AsyncSQLAlchemy 2.0	ACID transaction safety, row-level locking, automated migrations
Frontend UI	React 18, TypeScript, Vite 8, Tailwind CSS	Strict typing, hot module replacement, Material 3 aesthetic standards
Security & Auth	PyJWT (HS256), Passlib (Bcrypt), RBAC Guards	Role-based route isolation, database email verification enforcer
Artificial Intelligence	Google Gemini API (gemini-3.7-flash), RAG	Grounded HR policy QA, compensation explainability, workforce summaries
Quality Assurance	Pytest Async Suite, Pydantic v2 Schemas	100% automated contract verification, strict schema validation

3. Core Functional Modules & Innovations

Authentication & Access Lockdown

Supports multi-role registration (Employee, Admin/HR) with strict password security policy enforcement (8+ chars, uppercase, number, symbol) and a database-backed email verification enforcer preventing unauthorized sign-in.

Shift Attendance Management

Asynchronous clock-in/clock-out tracking with dynamic work hour calculations. Automatically classifies shift status into Present, Half-Day (< 4 hours), or Absent.

What-If Leave Simulator & Atomic Approvals

Provides employees with a real-time balance simulator to test dates prior to submission. Admin approval queue uses row-level database locking (with_for_update) for atomic balance deduction and mandatory reviewer feedback.

Payroll Calculation & Salary Slips

Calculates net compensation (Base + HRA + Allowances - Deductions). Provides read-only paystub access for employees and an admin payroll management dashboard with printable corporate salary slips.

Grounded Gemini AI Policy RAG Engine

Directly connects Google Gemini to structured DB models and verified corporate handbook files, delivering zero-hallucination policy answers, paystub explanations, and executive workforce metrics.

4. 5-Minute Hackathon Demo & Presentation Flow

Timestamp	Demonstration Focus	Screen Action & Core Talking Points
0:00 - 0:45	Registration & Security	Demonstrate Signup, password security policy checklist popover, and email verification lockout enforcer.
0:45 - 1:45	Employee Portal & Simulator	Clock in shift, navigate to Leave page, test What-If Leave Balance Simulator, and submit formal time-off request.
1:45 - 2:45	Grounded Gemini AI RAG	Query policy assistant regarding carry-forward leave rules. Highlight grounded responses without hallucinations.
2:45 - 4:00	Admin Command Center	Switch to HR Admin Arjun Rao. Review workforce analytics, approve leave with row-locking balance deduction, inspect payroll editor.
4:00 - 5:00	Notifications & Paystub Export	Switch back to Employee. Show real-time notification alert, open corporate Salary Slip modal, and deliver closing verdict.

5. Hackathon SRS Compliance Matrix

SRS Section	Requirement Description	Compliance Status
3.1 Auth & Verification	Sign Up, Sign In, Password Policy, Role Scoping, Email Verification	100% Implemented & Enforced
3.2 Dashboards	Role-Based Employee and Admin Executive Command Centers	100% Implemented & Live Data
3.3 Profile Management	Personal, Job, Salary Structure View & Limited/Full Edit Rights	100% Implemented
3.4 Attendance	Daily/Weekly Views, Check-In/Check-Out, 4-Status Tagging	100% Implemented
3.5 Leave Management	Leave Types (Paid/Sick/Unpaid), Date Ranges, Atomic Approvals & Rejections	100% Implemented
3.6 Payroll Control	Read-Only Paystubs, Admin Salary Editor, Printable Slips	100% Implemented
Section 6 Enhancements	Real-Time Notifications Engine, Analytics Dashboards, Gemini AI RAG	Exceeds Specification